

NISHA MAHEASVARAN

AI Data Analyst & AI Engineer • GenAI, Agentic Workflows & Analytics

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PROFESSIONAL SUMMARY

Strategic, results-driven **AI Data Analyst & AI Engineer** based in **Johor Bahru** with proven experience in end-to-end Python analytics (**pandas, numpy, scikit-learn**), SQL, and production-grade **Generative AI & Autonomous AI Agents**. Proven success designing and evaluating agentic workflows using **n8n, Langflow, and LangChain** with function calling to automate data querying, reporting, and operational tasks. Skilled in RAG retrieval from vector databases, business prompt engineering, and model governance (**token optimization, hallucination mitigation, and drift detection**). Certified Scrum Master skilled in translating business requirements into AI roadmaps and high-value solutions.

TECHNICAL SKILLS

Data Analytics & Extreme Python: Python (Advanced: *pandas, numpy, scikit-learn*), SQL (PostgreSQL, MySQL), Exploratory Data Analysis (EDA), Statistical Modeling, Feature Preparation, Automated Data Pipelines, Jupyter Notebooks

AI Agents & Workflow Automation: Autonomous AI Agents, Multi-Agent Systems, n8n, Langflow, LangChain, ComfyUI, Function Calling / Tool Calling, Task Automation (Alerts, Querying, Report Generation)

Generative AI, RAG & Model Evaluation: RAG Architectures (fastRAG, LangChain), Vector Databases (Chroma, FAISS), Prompt Engineering (Summarization, Q&A, Classification), Output Evaluation (Hallucination Mitigation, Accuracy, BLEU/ROUGE), LLaMA (2/3), Token Optimization

Data Pipelines, Backend & MLOps: Data Extraction & Transformation, FastAPI, RESTful APIs, Model Monitoring (Drift, Error Resolution), On-Premises Air-Gapped Deployment, Edge AI Inference, Docker, Linux, Git, CI/CD

AI-Augmented Dev & Methodologies: AI Coding Agents (Cursor, Antigravity, GitHub Copilot, Claude Code, OpenAI Codex, Kiro), Agile/Scrum (Certified Scrum Master, Sprint Planning), Stakeholder Strategy & AI Roadmapping

PROFESSIONAL EXPERIENCE

Imagine AI Sdn Bhd

📍 Johor Bahru, Malaysia

AI Engineer / AI Data Solutions Consultant (Part-Time)

Feb 2026 – Present

- Provide architectural advisory on enterprise data pipelines, model evaluation, and continuous platform stability.
- Spearhead comprehensive technical knowledge transfer and onboarding, mentoring incoming AI engineers on core system architectures, custom model weights, and pipeline maintenance.
- Advise on production inference tuning, latency debugging, and hardware acceleration across GPU, CPU, and NPU edge deployments.
- Maintain stability and reliability for continuous production deployments of the on-premises AI Teacher Bot platform.

AI Engineer / AI Data Analyst

Jan 2025 – Feb 2026

- Architected and deployed commercial-grade **Generative AI and RAG solutions**, preparing internal structured data in vector databases to serve low-latency business Q&A and summarization pipelines.
- Engineered enterprise **Autonomous AI Agents** and automated workflows using **n8n, Langflow, and LangChain** with custom function calling, automating data extraction, scheduled reporting, and alerting.
- Built end-to-end data processing pipelines and analytical notebooks in **Python (pandas, numpy, SQL)**, transforming multi-source enterprise data into structured datasets for model evaluation and downstream agentic use.
- Tracked and evaluated production model performance, establishing benchmarks for **accuracy, hallucination reduction, and token optimization**, proactively resolving errors and drift across client deployments.
- Delivered cross-platform AI solutions for enterprise clients including **Dell, Lenovo, Huawei, Honor, and Intel**, optimizing inference performance across CPU, GPU, and NPU architectures.
- Served as **Scrum Master** and technical lead for 4 client engagements, translating business requirements into AI product roadmaps and clear value metrics while mentoring junior developers.
- Engineered real-time **AI Avatar systems** combining speech synthesis with lip-syncing (**Wav2Lip, OpenVINO**) and developed asynchronous backend APIs using **FastAPI**.

AI Developer Intern

Sep 2024 – Dec 2024

- Researched and prototyped natural language processing (NLP) and computer vision modules for real-world enterprise applications.
- Explored parameter-efficient fine-tuning (PEFT/LoRA) and model quantization strategies to optimize inference speed on compute-constrained edge hardware.
- Collaborated with senior engineering staff to benchmark deep learning pipelines and transition experimental prototypes into production-ready scripts.

KEY PROJECTS & TECHNICAL COLLABORATIONS

Live Avatar Chatbot with Intel | *Intel fastRAG, OpenVINO, Wav2Lip, Python, FastAPI*

On-Premises / Enterprise

- Engineered an end-to-end on-premises interactive avatar pipeline capable of real-time conversational responses with Intel-optimized hardware acceleration.
- Integrated **Intel fastRAG** with **OpenVINO** and **Wav2Lip** to streamline audio synthesis and facial animation, reducing response latency so avatars speak concurrently with ongoing token generation.
- Packaged the complete solution into a robust software module delivered to frontend and product teams for customer commercialization; designated technical lead overseeing on-premises and cloud robotic chatbot solutions.

Diffusion Models for Text-to-Video Generation | *Flux, Hunyuan, ComfyUI, Python*

Industry Collaboration

- Directed the technical design of an automated Text-to-Video generation pipeline utilizing state-of-the-art diffusion models (Flux, Hunyuan).
- Built modular prompt-processing workflows in **ComfyUI**, integrating semantic guards and context-aware styling to produce high-fidelity video clips in collaboration with **Bitsmedia**.

Persona Clones via LLaMA 2 Fine-Tuning | *LLaMA 2, Hugging Face, LoRA, QAT, BLEU, ROUGE*

Open-Source / Research

- Curated, cleaned, and tokenized custom conversational datasets to fine-tune LLaMA 2 for highly personalized conversational persona emulation.
- Implemented transfer learning pipelines with **LoRA** and **Quantization-Aware Training (QAT)**, evaluating output quality and persona consistency using standard linguistic metrics (**BLEU, ROUGE**).

Industrial Vision & Automated Coin Counting | *YOLOv8, OpenCV, Custom CNNs*

Industry Collaborations

- Collaborated with manufacturing partner **TSH Company** to develop automated object detection for industrial operational environments, validating performance across harsh factory lighting.
- Built an automated real-time currency recognition and calculation system using custom-trained **YOLO** architectures with augmented coin datasets.

EDUCATION

Raffles University

Medini, Iskandar Puteri, Malaysia

Bachelor of Information Systems (Hons) in Artificial Intelligence

Graduated

Key Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms, Statistical Methods for AI, Big Data Analytics

PROFESSIONAL CERTIFICATIONS

- Building Cyber Resilience in a Digital Era – Cybersecurity:** National University of Singapore (NUS)
Credential ID: 169721679 | Comprehensive assessment of corporate infrastructure defense, threat mitigation, and data resilience.
- Software and Apps Development with C#:** Koto Association

Dec 2025
Dec 2025

LEADERSHIP, MENTORSHIP & COMMUNITY ENGAGEMENT

- Treasurer, AI Club (Raffles University):** Managed organizational finances, coordinated technical seminars, and led student cohorts in executing AI-for-good prototype hackathons (Jan 2023 – Nov 2023).
- Keynote Presenter, AI Lab Inauguration:** Selected by university faculty to demonstrate flagship AI research and live neural network prototypes at the official Raffles University AI Lab opening ceremony.
- Technical Mentor & Instructor:** Guided junior developers and interns across Python software architecture, deep learning fundamentals, and computer vision; monitored and mentored trainees within the regional K-Youth programme.
- Community Workshops & Hackathons:** Conducted hands-on technical workshops on Transformers, LLMs, and OpenCV for developer communities; active open-source contributor on GitHub and participant in multi-day hackathons (2022, 2023).

PROFESSIONAL REFERENCES

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